

Generation Interconnection System Impact Study Report

For Queue Project Q17-0834

Farmville 138 kV

January 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0834. The Interconnection Customer proposes a 125.4 MW wind facility interconnecting at the Farmville 138 kV substation in the Virginia area. The request was received on July 22, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 313 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$5,915,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Farmville – Wheelabrator S Broward 138 kV

Contingency: Bear – Leavitt 345 kV Line 1

Loading: 119.6% (Rating: 259 MVA)

DFAX: 26.7%

Pre-project loading: 106.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Farmville – Wheelabrator South Broward Substation 138 kV L	Rebuild line with higher-capacity conductor	\$7,531,000	78% (\$5,915,000)
	Total allocated to project		\$5,915,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.